CASE III - Reversible Data Perturbation for Privacy Preservation in IIoT Systems: Evaluating the Privacy–Utility Trade-off and Identifying the Most Efficient Method

Group Members:

Name	ID
Rabbi Hasan Himel	20251006057
Tayyaba Basri	20251006032
Hamza Haroon	20251006041

RESEARCH PURPOSES

Defending Against Industrial Espionage: A Reversible Perturbation Approach for Preserving Trade Secrets in IIoT Telemetry

Dataset

Link:
https://research.unsw.edu.au/projects/toniot-datasets
(Dataset created by University of New South Wales, Australia)

File: Train_Test_IoT_Modbus.csv

Nour Moustafa > TON_IoT datasets > Train_Test_datasets > **Train_Test_IoT_dataset**

Research Purpose: Securing the Industrial Blueprint

The Modbus protocol serves as the backbone of critical infrastructure, governing communication between PLCs and sensors in sectors ranging from manufacturing to energy grids. However, Modbus was designed without native security or encryption. In the modern IIoT era, this creates a vulnerability: **Operational Espionage**.

The raw telemetry transmitted via Modbus temperature setpoints, valve pressures, motor speeds constitutes the intellectual property of industrial processes. If intercepted, competitors can reverse-engineer manufacturing recipes, and attackers can analyze baselines to orchestrate targeted physical attacks.

Standard encryption is often too computationally heavy for legacy PLCs and prevents real-time cloud analytics. This research applies **Reversible Data Perturbation** to Modbus registers, disguising production logic as noise during transmission. Trusted cloud systems can reverse the noise for predictive maintenance.
In IIoT, "Privacy" means hiding the Production Recipe.

Modbus is the language of Operational Technology (OT).
It controls oil pipelines, water treatment plants, and automotive assembly lines.

Example:

- *Scenario:* A chemical plant producing a patented medicine
- *Data:* “Maintain Reaction Tank at 85.5°C”, “Mixer Speed 1200 RPM”
- *Leak:* Competitors can steal the manufacturing formula
- *Solution:* Perturbation masks these values (e.g., 85.5 → 214.3)

Lightweight Method: In the context of IIoT (Industrial IoT), **Lightweight** refers to an algorithm that is mathematically simple enough to run on very weak hardware (like a sensor or a 20-year-old PLC) without draining its battery or causing delays.

Think of it like this:

- **Heavyweight Method (e.g., AES Encryption, Homomorphic Encryption):** This is like putting the data in a heavy steel safe. It is very secure, but it takes a lot of time and effort (CPU cycles) to lock and unlock it. If a sensor needs to send data every 1 millisecond, the "locking" process is too slow.
- **Lightweight Method (e.g., Reversible Perturbation):** This is like writing the data in a "secret code" using simple math (addition/multiplication). It is incredibly fast and uses almost no battery. It is secure *enough* to confuse a spy, but fast enough for real-time manufacturing.

Why It Is Concerning (The Threat)

1. **Economic Loss (Espionage):** Competitors can determine:

   a) robot arm movements
   b) yield rate
   c) throughput
   d) cycle times

2. **Targeted Attacks (Sabotage):** Attackers watching Modbus traffic can learn:

   - normal valve pressure ranges
   - machine operation patterns

This enables stealthy attacks that appear normal.

3. The Leak: If a competitor intercepts this unencrypted Modbus traffic, they don't just see "data." They see the exact manufacturing formula. They can steal the intellectual property without ever entering in the building.

The TON_IoT Modbus dataset represents real-time industrial telemetry that reveals operational cycles and process behavior, making reversible perturbation essential to stop industrial espionage

while enabling anomaly detection. “Lightweight” means the method can run on IIoT edge devices without slowing the production line.

Recent Works

1. Privacy Preservation for Machine Learning in IIoT Data via Manifold Learning and Elementary Row Operations, Yetkin, E. Fatih; Ballı, Tuğçe; International Conference on Information Systems Security and Privacy, (2025) Vol.2, 607–614. DOI: 10.5220/0013275000003899
2. Alsaedi, Abdullah, Nour Moustafa, Zahir Tari, Abdun Mahmood, and Adnan Anwar. “TON_IoT telemetry dataset: a new generation dataset of IoT and IIoT for data-driven Intrusion Detection Systems.” IEEE Access 8 (2020): 165130–165150.
3. Privacy Preservation for Machine Learning in IIoT Data via Manifold Learning and Elementary Row Operations, Yetkin, E. Fatih; Ballı, Tuğçe; International Conference on Information Systems Security and Privacy, (2025) Vol.2, 607–614. DOI: 10.5220/0013275000003899
4. A Hybrid Approach With GAN and DP for Privacy Preservation of IIoT Data, Hindistan, Yavuz Selim; Fatih Yetkin, E.; IEEE Access, (2023) Vol.11, 5837–5849. DOI: 10.1109/ACCESS.2023.3235969
5. Hartog. “ToN IoT—The role of heterogeneity and the need for standardization of features and attack types in IoT network intrusion datasets.” IEEE Internet of Things Journal (2021).
6. Moustafa, Nour, M. Ahmed and S. Ahmed. “Data Analytics-Enabled Intrusion Detection: Evaluations of ToN_IoT Linux Datasets.” TrustCom 2020.
7. Moustafa, Nour. “New Generations of Internet of Things Datasets for Cybersecurity Applications based Machine Learning: TON_IoT Datasets.” eResearch Australasia Conference, 2019.